Math F404: Homework 1 Due: September 8, 2026

PROBLEMS 0.1.

PROBLEM 1. Sutherland 4.1

PROBLEM 2. Sutherland 4.2

PROBLEM 3. **(AI Free)**

Let $A \subset \mathbb{R}$ be a nonempty set that is bounded above. Let $m = \sup A$. Show that for every $\epsilon > 0$ there exists $a \in A$ with $m - \epsilon < a \leq m$.

PROBLEM 4. Suppose (x_n) is a sequence converging to a limit x.

Show that (x_{n+1}) converges to the same limit.

Show that if $c \in \mathbb{R}$ then (cx_n) converges to cx.

PROBLEM 5. Suppose $a \in \mathbb{R}$ and suppose $a^n \to L$. Use the previous problem to show that either $L = 0$ or $a = 1$ (in which case $L = 1$).

PROBLEM 6. Suppose $0 < a < 1$. Use the monotone convergence theorem to show $a^n \to 0$. Hint: the previous problem could be helpful.

PROBLEM 7.

Knowing that $\sqrt{2}$ is irrational, show that $q\sqrt{2}$ is irrational for all $q \in \mathbb{Q}$.

Sutherland 4.8. **Hint:** consider $a/\sqrt{2}$ and $b/\sqrt{2}$ as endpoints first.

PROBLEM 8. Show directly from the definition that $x_n = (-1)^n$ is not Cauchy and conclude that it does not converge.

PROBLEM 9. Sutherland 4.12 **Hint:** For simplicity, assume f and g are defined on all of $\mathbb{R}$, and take f to be really boring.

PROBLEM 10. AI Use. Please describe how you used AI to help complete this assignment.

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